

Nonrenewable Resource Extraction with Salvage Value and Varied Discount Rates: A Case for Crude Oil

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Introduction & Background

The objective of this project is to show how the extraction behaviors of monopolists who own or have rights over nonrenewable resources (in this case, crude oil) change in the presence of salvage values and how changes in discount rates might affect the optimal extraction paths for such firms. There have been numerous studies on how firms might be incentivized or forced to not exhaust their disposable resources, and the introduction of a salvage value is one such case where it might happen. In this paper, salvage is introduced in the form of an endowment that the extracting firm receives from the government. Before starting extraction, the government offers the firm a contract that guarantees it a fixed payment of a certain amount (for research purposes), that is competitive with the market dynamics at that time, if it agrees to stop extraction after a certain level. As the firm approaches that prespecified milestone, it compares the current market conditions with the government's payment and decides to stop extraction at that level if the offer outweighs the company's profit prospects in the market. On the other hand, if the discount rate for the resource goes up (compared to other financial or natural assets and investment prospects), this should provide the firm with an incentive to capitalize and extract more aggressively and reach the stopping point earlier.

Some of the studies that examined relevant ideas include Wang and Zhao (2013)ⁱ where the researchers presented a case for a monopolistic firm that owns a nonrenewable resource and competes with other renewable firms in the market in terms of price. As the price of the nonrenewable resource gets close to the cost of renewable energy, the monopolist might increase their extraction and flood the market with their nonrenewable resource, keeping the prices low and making it difficult for renewable energy to compete. However, when the monopolist can't continue to operate economically by artificially keeping the prices down for their products, it might stop extracting or even consider investing in renewable energy at that point.

On the other hand, Nyambuu and Semmler (2014)ⁱⁱ discussed about how differing interest rates might influence the optimal extraction of resources. They mentioned that Hotelling originally assumed the prices for a competitive firm to be constant; however, this is limiting as most nonrenewable resource owners are mostly monopolists and their prices are reactive to changes in the discount rates. On a similar note, Cherian, Patel, and Khripko (1998)ⁱⁱⁱ mentioned that the

change in discount rates resulting from uncertainty in prices can affect the extraction behavior of firms, potentially leading them to stop extraction with some reserves left, which further technological innovations can make economically viable in the future. Earlier, Krautkraemer (1989)^{iv} also reiterated the same sentiment that firms may respond to incentives to stop extracting if the efforts of extracting at a considerable depth provide them with enough technological limitations, even if the market dynamics make the venture economically profitable.

Echoing these studies, Yang and Davis (2018)^v, in their research, talked about financial incentives, such as Pigouvian stock taxes, impacting non-renewable resource extraction after a certain level and changing interest rates potentially affecting the present value of future costs associated with pollution cleanup, indirectly shaping incentives for timely remediation and sustainable resource management under different regulatory frameworks. Growth and Schour (2006)^{vi} also argued that a progressive tax might encourage conservation efforts, leading to a higher salvage value for resources remaining in the ground. The latter of the two researchers, Schou(2000)^{vii}, also talked about how the value of environmental preservation (or the cost associated with it) after a certain level of extraction activity could influence a firm to stop extraction before exhaustion, which essentially acts as a salvage for the firm, in one of his earlier studies.

The oil and gas industry's use of sophisticated extraction techniques, such as lateral drilling and hydraulic fracturing, has attracted increased government attention and research investments in recent years. This is primarily due to growing concerns about the environmental consequences of traditional coal mining practices ^{viii}. Hydraulic fracturing, a technique used to extract oil and gas from rock formations, cannot be used for coal since it is extracted in solid forms ^{ix}. As a result, both the public and private sectors have placed increasing emphasis on research and development investments for crude oil and natural gas over the last couple of decades.

However, with an expedited extraction rate due to fracking, the quality of the deeper reserves declines faster over time. These reserves, located below the top layers, have high levels of sulfur and are thick in consistency, which is undesirable for both extraction companies and refiners ^x. By implementing a government investment strategy, like the one described earlier, where the government purchases the reserve once a certain level of extraction has been reached, producers would be motivated to refrain from extracting the remaining reserve that lacks economic viability for them. Instead, they would choose to leave the remaining oil untouched and benefit from

receiving the endowment, thereby maximizing their dynamic returns from the mine. In exchange, the government may undertake research on sour crude to promote and develop technology that could lead to improved methods that could make the extraction of the remaining reserves economically feasible, benefiting both the company and society as a whole. This provided an impetus for the study.

Therefore, it is clear from the previous discussion that although numerous studies have examined the extraction behaviors of non-replenishable resources in relation to market disincentives or regulation, none of them have specifically addressed the idea of providing extracting firms with an incentive, such as the government buying out the mine (or purchasing the right to the rig), to cease extraction without depleting the resources. This possibility seems economically viable, especially given the growing popularity of fracking technologies. This paper aims to fill in that gap by addressing the subject matter and showing how the extraction rate, depletion rate, growth of shadow value, prices, and the current value Hamiltonian change over time in a dynamic setting in the presence of a shadow value.

Model

For this model, the objective functional is,

$$\text{Max } J = \int_0^T (p(u)u - c(u))e^{-rt} dt + S_{x(T)} e^{-rt}$$

subject to $\dot{x}(t) = -u(t)$, $x(0) = x_0$, $x(t) \geq 0$, $S_{x(T)} = \psi(x(T)) = SxT$, $S < a$, the choke price

where u is the control variable (nonlinear in the objective functional) representing the extraction rate, $x(t)$ is the reserve stock, $xT = x(T)$ is the final remaining reserve, $p(u)$ is the demand function where $p(u) = a - bu$, and c is the cost of extraction where $c(u) = \frac{u^2}{2}$. $S_{x(T)}$ here represents the salvage value, and a , b , r , and $S > 0$.

This model assumes a nonlinear control mechanism that allows the monopolist to adjust extraction rates in response to various factors, such as uncertainties, technological constraints, market conditions, regulatory policies, resource dynamics, variations in resource quality, and the possible emergence of alternative fuel sources. Similarly, it is considered that the cost function is nonlinearly growing because, as the quality of the resource diminishes and the depth increases over time due to extraction, the cost of extraction also increases (Livernois, 1987)^{xi}.

Additionally, the discount rate (opportunity cost of investing in this endeavor) in this case is assumed to be set by finding the average yields from corporate bonds, risk-free government bonds (like AA utility bonds), and investments in technological and financial assets, while also accounting for the current state of the market and historical discount rates from the natural assets market.

So, the current value Hamiltonian is,

$$\mathfrak{H} = F + \mu f = (p(u)u - c(u)) + \mu \dot{x}(t) = (a - bu)u - \frac{u^2}{2} - \mu u,$$

where μ is the current value of the shadow price.

The Maximum Principle

(i) The First Order Condition (FOC),

$$\mathcal{H}_u = 0 \Rightarrow \mu = p'(u)u + p(u) - c'(u) = a - (2b + 1)u$$

says that the monopolist should continue extracting oil until the marginal revenue received from selling an additional unit, $a - (2b + 1)u$, just balances the foregone value of an additional (in situ) unit of oil not extracted (μ). So μ here is the marginal value of preserving the resource. In other words, when this equation is satisfied, the marginal contribution of the stock is just equal to the difference of marginal revenue and marginal operating costs (operating profit).

Here, u and μ are negatively related, and so as μ increases, u falls (if the value from preservation goes up at a given time, the firm would want less extraction and more preservation at that time) and vice versa. In other words, when the opportunity cost of extraction operations goes up, extraction activities go down.

If $\mu < a - (2b + 1)u$ at some point in time, the producer has an incentive to increase extraction and earn more rent until the inequality returns to equality when the producer becomes indifferent between present and future returns.

On the contrary, if $\mu > a - (2b + 1)u$ holds, the total cost of extraction becomes greater than the benefits of doing so, and the firm cuts back on extraction (marginal benefit from preservation $>$ marginal benefit from extraction) until the equality is restored, where it has no incentive to deviate anymore.

(ii) The Dynamic Constraint (DC),

$$\dot{x}(t) = -u(t), x(0) = x_0, x(t) \geq 0$$

here suggests that the rate of change of the remaining reserve is equal to the negative of the extraction rate, $u(t)$. So, as extraction increases, the remaining reserve diminishes. It also says that the total oil reserve in the mine (initially) is a constant x_0 , which doesn't replenish on itself, and that the amount of reserve is nonnegative at all times.

(iii) The Portfolio Balance (PB) here,

$$\dot{\mu}(t) = r\mu - \mathcal{H}_x$$

contains 3 components. The left side of the equation features the term $\dot{\mu}(t)$, which indicates how fast the shadow price (marginal contribution of the oil stock) is changing over time while on the right side, $r\mu$ refers to the intrinsic growth (regardless of extraction) of the shadow price (value of preservation) through time, and $\mathcal{H}_x$ represents the marginal cost of keeping the resource in the reservoir (as stock increases, extraction and therefore sales profit goes down, pushing the value of current value Hamiltonian down with it).

So, if $r\mu > \mathcal{H}_x$ overtime, the marginal value of preservation goes up through time and vice versa. Here, however, $\mathcal{H}$ is not a function of x , and so here, $\dot{\mu}(t) = r\mu$.

This equation can be expressed as $\frac{\dot{\mu}}{\mu} = r$, where $\frac{\dot{\mu}}{\mu}$ is the growth rate of net marginal benefits from preserving the oil (own rate of return or capital gain) and this equation (following Hotelling's rule) says that at the optimal extraction path u^* , the net marginal benefit from holding one barrel of oil unextracted in situ equals the required rate of return.

If $\frac{\dot{\mu}}{\mu} > r$ at time t , the future returns are valued more than the present returns, and the firm reduces its extraction at time t until the equality is achieved. In other words, when the own rate of return is greater than the required rate of return, the firm chooses preservation over extraction.

On the other hand, if $\frac{\dot{\mu}}{\mu} < r$ at time t , the future returns are valued less compared to the present returns, and the firm increases its extraction at time t until the equality is achieved. In other words, when the own rate of return is less than the required rate of return, the firm chooses extraction over preservation.

Only at $\frac{\dot{\mu}}{\mu} = r$ equality is the firm indifferent between the future and present returns (dynamically arbitrated) and decides to extract optimally on this path.

The Optimal Paths

Solving the Portfolio Balance $\dot{\mu}(t) = r\mu$, gives

$$\mu^*(t) = Ce^{rt}$$

by solving the differential equation, $\dot{\mu}(t) - r\mu = 0$ using the formula for first order linear differential equation of the form, $z' + R(t)z = Q(t) \Rightarrow z = CI^{-1} + \frac{Q}{R}$.

Here, $R(t) = -r$, integrating factor, $I(t) = e^{\int R(t) dt} = e^{\int -r dt} = e^{-rt}$, $Q(t) = 0$, and hence for a time invariant Q , $\mu^*(t) = CI^{-1} + \frac{Q}{R} = Ce^{rt}$, where C is the constant of integration.

If $C < 0$, then the marginal benefit of preservation, $\mu(t)$ would be negative, and there would be too much extraction as the firm would have no incentive to preserve anything for the future, which is a trivial case. Similarly, if $C = 0$, $\mu(t) = 0$, and there would be no benefit from holding on to the reserve or otherwise, and the firm would extract everything now (myopic case) to meet their contractual/market obligations or fixed costs which would turn this problem into a static one. For a dynamic solution, $C > 0$ needs to be satisfied, in which case $\mu(t)$ will be positive and it will grow exponentially until $\mu(T) = S$, the salvage value.

When μ hits this scrap value, the firm stops their extraction to capitalize on this value, which in this case is the money they expect to receive from the government for research purposes. Solving $\mu(t) = Ce^{rt}$ with this boundary condition gives $Ce^{rT} = S \Rightarrow C = Se^{-rT}$.

This gives the firm its explicit solution for the shadow price,

$$\mu^*(t) = Se^{r(t-T)}$$

This optimal shadow price here negatively depends on the discount rate (since $t \leq T$) since a higher discount rate (interest rate) indicates that the firm could extract now and earn more interest payments on its sales, and not doing so would increase its opportunity cost. As a result, a higher discount rate implies that the firm values the present more than the future (more extraction now than the future), and vice versa.

Plugging this value of $\mu^*(t)$ into the FOC,

$$a - (2b + 1)u = Se^{r(t-T)}$$

$$\Rightarrow u^*(t) = \frac{1}{2b+1} [(a - Se^{r(t-T)})]$$

where $u^*(t)$ is the optimal extraction rate over the time horizon.

Now, totally differentiating the First Order Condition with respect to t :

$$-2b\dot{u} - \dot{u} = \dot{\mu} = r\mu \Rightarrow (2b + 1)\dot{u} = -r(a - 2bu - u) \Rightarrow \dot{u} = -r \frac{a - 2bu - u}{2b + 1}$$

where the ' $a - 2bu - u$ ' term in the numerator represents rent, which, for all practical purposes, is positive; r is the discount, which is also positive for nontrivial cases, and ' $2b + 1$ ' in the denominator is positive as b is assumed to be a positive constant in the model. This makes the right-hand side negative, implying that the rate of extraction falls over time (albeit at different paces for different discount rates). However, since $u(t) \geq 0$ and $\dot{x}(t) = -u(t)$ from the dynamic constraint, the reserve also falls overtime.

Now, for the first order linear differential equation,

$$\dot{u} = -r \frac{a - 2bu - u}{2b + 1} \Rightarrow \dot{u} - ru = -\frac{ra}{2b + 1}$$

where the integrating factor, $I(t) = e^{\int R(t) dt} = e^{\int -r dt} = e^{-rt}$, $Q(t) = -\frac{ra}{2b + 1}$; following the method for solving the first order linear differential equations,

$$u^*(t) = I^{-1}(t) [\int I(t)Q(t)dt + L] = e^{rt} [e^{-rt} (-\frac{ra}{2b + 1}) + L] = e^{rt} [\frac{ae^{-rt}}{2b + 1} + L] = Le^{rt} + \frac{a}{2b + 1}$$

where L is the constant of integration. Now, from FOC,

$$\Rightarrow u(T) = -\frac{\mu(T) - a}{(2b + 1)} = \frac{a - S}{(2b + 1)} = \frac{a}{(2b + 1)} - \frac{S}{(2b + 1)} \quad \{\text{using } u(T) = S\}$$

Using it into the above equation,

$$\Rightarrow Le^{rT} + \frac{a}{2b + 1} = \frac{a}{(2b + 1)} - \frac{S}{(2b + 1)} \Rightarrow L = -\frac{Se^{-rT}}{(2b + 1)}$$

And this leads to the equation,

$$\Rightarrow u^*(t) = \frac{a}{2b+1} - \frac{Se^{-rT}}{(2b+1)} e^{rt} = \frac{1}{2b+1} [(a - Se^{r(t-T)})]$$

Which is the same expression found above for $u^*(t)$.

Here, for any $S < a$, which is the price threshold at which extraction becomes economically unviable and undesirable, $u^*(t)$ is always positive, and as t goes from 0 to T , $Se^{r(t-T)}$ grows bigger in value and the extraction rate gradually falls until $Se^{r(t-T)}$ equals S at time T , where $u(t)$ shrinks to the terminal level, $u^*(T) = \frac{a-S}{2b+1}$.

So, as long as $S < a$, the extraction rate never goes to zero. The assumption $S < a$ here ensures that the firm reaches its salvage value before reaching the choke (cutoff) price (where the demand goes to zero) and it optimizes its returns by capturing the salvage value, by stopping extraction before exhausting the resources, which would not be possible if $S \geq a$ and the resources were exhausted before reaching the salvage value.

This assumption can also be confirmed analytically from the transversality condition,

$$(\mu(T) - S)(x(T) - k) = 0$$

At the terminal time, as $\mu(T)$ equals S , $x(T)$ equals k where $k > 0$ and so $x(T) > 0$. This indicates that the extraction activities stop before the resource is fully exhausted and $u^*(T) > 0$.

Also, using the Free time transversality condition (FTTC),

$$\mathfrak{H}(x(T), u(T), \mu(T), T) - rSx(T) = 0$$

$$\Rightarrow (a - bu(T))u(T) - \frac{u(T)^2}{2} - \mu(T)u(T) - rSxT = 0$$

$$\Rightarrow xT = \frac{(a - bu(T))u(T) - \frac{u(T)^2}{2} - \mu(T)u(T)}{rS} = k > 0$$

which leads to the same conclusion of non-exhaustion.

Dividing the FTTC by u ,

$$\frac{u(T)}{2} = a - bu(T) - \mu(T) - \frac{rSxT}{u(T)}$$

where $\frac{u(T)}{2}$ is the average cost.

However, from FOC at T,

$$u(T) = a - 2bu(T) - \mu(T)$$

where $u(T)$ is the marginal cost.

Setting the average and marginal costs equal to each other,

$$a - bu(T) - \mu(T) - \frac{rSxT}{u(T)} = a - 2bu(T) - \mu(T)$$

and simplifying this equation, it is found that,

$$u^*(T) = \sqrt{\left(\frac{rSxT}{b}\right)}$$

where r , S , xT , and b are all positive and hence $u^*(T)$ is also positive (non-exhaustion).

In the case of exhaustion, $u(T)$ should be 0 when the marginal and the average cost balance out. However, when the marginal cost is set against the average cost in this case, it is found that $u^*(T) > 0$, indicating that the extraction stops before exhaustion.

Now, solving the differential equation from the Dynamic Constraint, $\dot{x}(t) = -u(t)$ on Mathematica,

$$x^*(t) = \frac{S - e^{-r(t-T)}S - ar(t-T)}{(1+2b)r}$$

Where the stock is a function of the discount rate, time, salvage value, and the constants 'a' and 'b'.

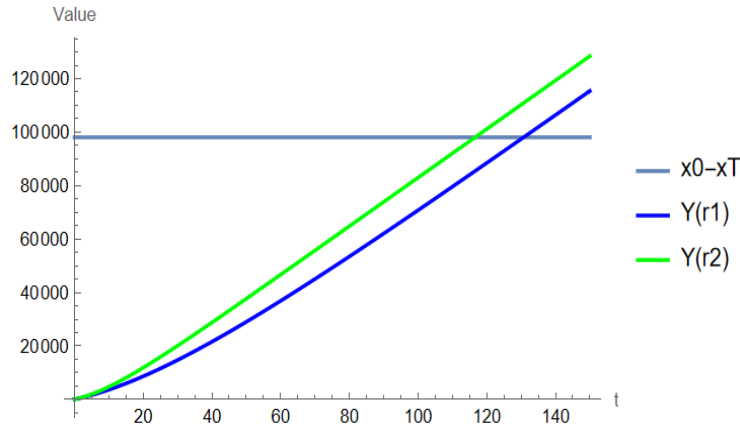
Now, in order to find the optimal T, the equation $x_0 - x_T = \int_0^T udt$ is solved since the difference between the initial and final reserve amount can be traced back by integrating the total extraction over the time period $[0, T]$. Here, x_T is the final reserve level, $x(T)$. Solving this integration using Mathematica,

$$x_0 - x_T = \int_0^T udt = \int_0^T \frac{1}{2b+1} [(a - Se^{r(t-T)})]dt,$$

the following result is found:

$$x_0 - x_T = \frac{e^{-rT}S + arT - S}{(1+2b)r}$$

Now, using the values $a = 1000$, $b = 0.05$, $x_0 = 100000$, $S = 700$, and two distinct discount rates $r_1 = 3\%$ and $r_2 = 8\%$, the following figure is plotted on Mathematica.



$T_1(r_1) : 130.727$

$T_2(r_2) : 116.605$

Using the aforementioned values, x_T is numerically solved by Mathematica which gives the value $x_T \approx 1949$. Using x_T in $x_0 - x_T = \frac{arT + e^{-rT}S - S}{r + 2br}$, two distinct optimal terminal times, T_1 and T_2 , are found for their corresponding discount rates (r_1 and r_2) where $T_1^* = 130.727$ and $T_2^* = 116.605$.

With these values, $x^*(t)$, $u^*(t)$, $\mu^*(t)$, $\mathcal{H}^*(t)$, $P^*(t)$, $P^*(u)$ can also be plotted for both T_1 and T_2 (and the corresponding r_1 and r_2).

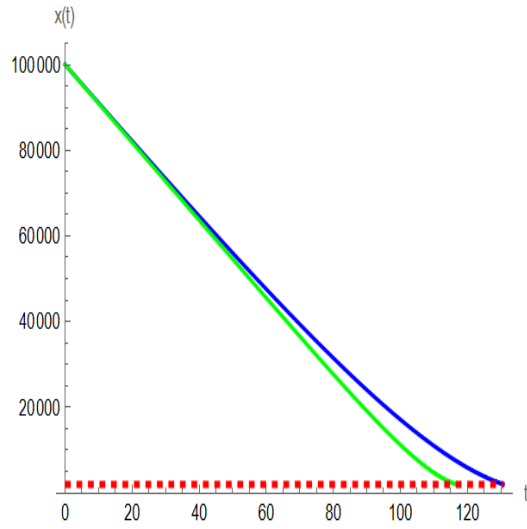


Figure 1: $x^*(t)$

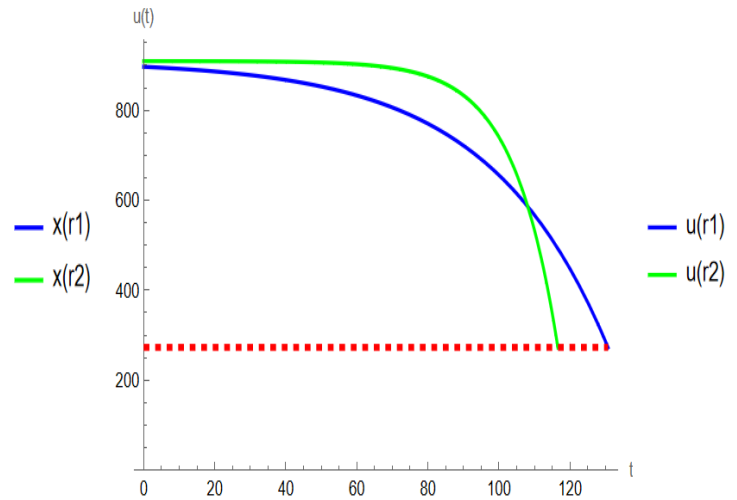


Figure 2: $u^*(t)$

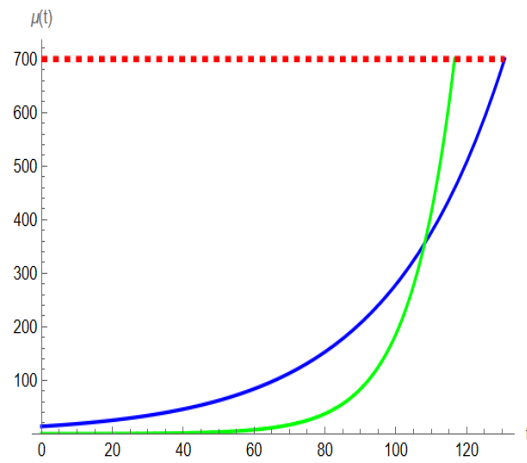


Figure 3: $\mu^*(t)$

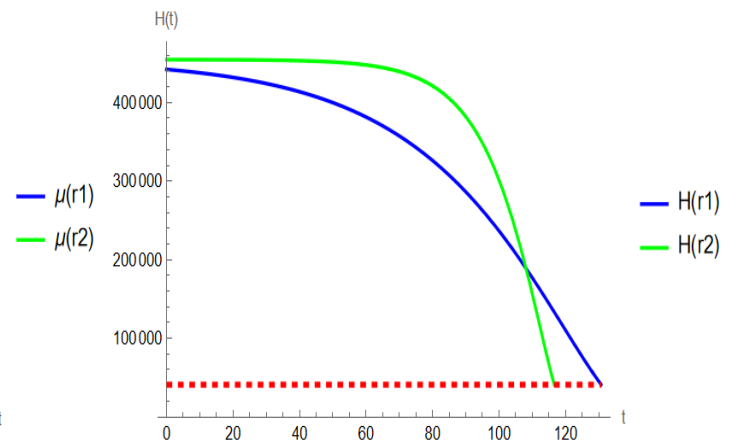


Figure 4: $\mathcal{H}^*(t)$

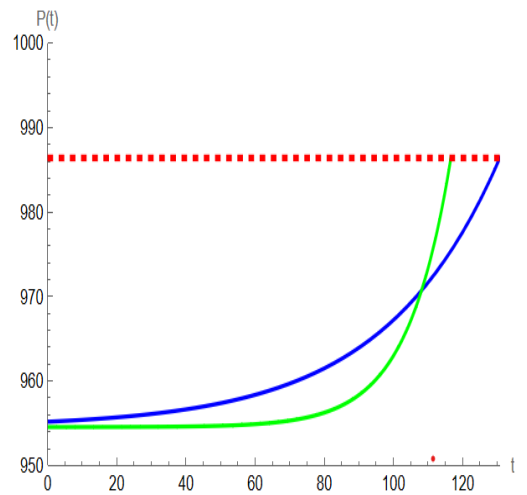


Figure 5: $P^*(t)$

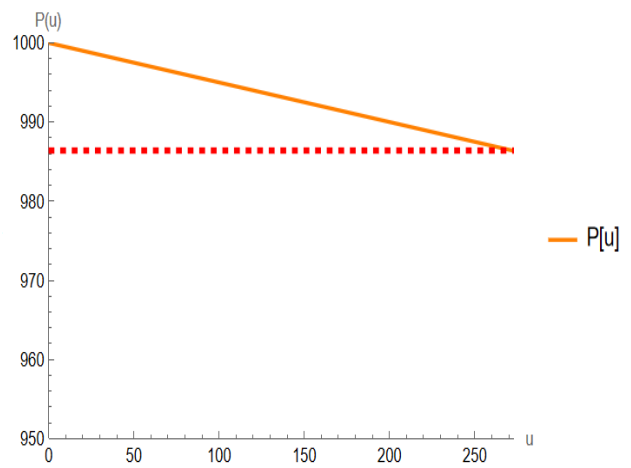


Figure 6: $P^*(u)$

Discussion & Recommendation

T_1 (r_1) being greater in value than T_2 (r_2) follows economic rationale as a bigger discount rate would expedite the extraction rate, making the firm reach the final extraction rate $u(T)$ faster and thereby reaching the final remaining reserve level xT quicker as it reaches $\mu(T) = S$ in a shorter time. Also, in the presence of the salvage value, $\mathcal{H}^*(t)$ never reaches zero as $u^*(t)$ never reaches zero and $p^*(t)$, and $\mu^*(t)$ never reaches 'a', the maximum price. This reflects that the presence of the salvage value influences the firm's extraction behavior, leading it to a strategic balance between maximizing current profits and capitalizing on future benefits from salvage (maximizing its overall value over time).

The figures above show that the firm ceases its mining operations when there are $x^*(T) = 1949$ barrels of oil remaining in the ground. When $\mu^*(T)$ reaches $S = 700$, the extraction stops with $u^*(T) = 272.73$, $\mathcal{H}^*(T) = 40909.09$, and $p^*(T) = 986.36$ and the discount rate only influences the rate (path) of reaching the final level rather the final level itself. Another noteworthy observation is how $\mu^*(t)$ and $p^*(t)$ produce very similar-looking figures, and $u^*(t)$ (and $\mathcal{H}^*(t)$) is an inverse of them, which is aligned with the model because from FOC, it is seen that $\mu(t)$ (and therefore $p(t)$) has a negative relationship with $u(t)$. This is because price P (and therefore μ), from demand side, has a negative relationship with quantity (extraction u), something that was also resonated by the empirical work of Wang and Zhao (2013)ⁱ in the context of firms saturating the market with more resources to artificially keep prices down. The $P(u)$ figure from above also illustrates this. Hence, as the extraction rate falls over the time horizon, the price increases.

Additionally, as the discount rate increases from 3% to 8%, the extraction rate initially jumps up vertically as the present benefit of extraction becomes more attractive relative to the future, and with a higher return at every time period t , $u(r_2)$ not only starts from a higher spot but also reaches the $u(T)$ level sooner than $u(r_1)$. Since $\mathcal{H}^*(t)$ represents the value received from extraction, it takes on an analogous course. The corresponding mirrored images can be noticed in $\mu(t)$ and $p(t)$ as they start below their initial levels with the extraction rate going up (more extraction leading to lower prices and a higher interest rate leading to a reduced attraction for the prospect of preservation) and then reaching $\mu(T)$ and $p(T)$ for r_2 sooner compared to the r_1 level. Unsurprisingly, $x(t)$ also

generates a steeper figure for r_2 , as the resource is depleted faster than at the r_1 level, reaching $x(T)$ sooner.

If, on the other hand, the salvage value was to increase from S to S' where $S' < a$ still holds, the firm would cease its extraction activities relatively later as it would take longer for its shadow value to reach the S' level compared to the S level. This is because, in order to reach a higher level of salvage, the firm would have to extract more, resulting in sufficiently more scarcity for the oil deposit in the reservoir, which would further increase the value of the marginal contribution of the remaining resource to the higher level of S . As a result, T' would be higher than T^* , $\mu'(T)$ would be greater than $\mu^*(T)$, $p'(T)$ would be greater than $p^*(T)$, $u'(T)$ would be less than $u^*(T)$, $\mathcal{H}'(T)$ would be less than $\mathcal{H}^*(T)$, $x'(T)$ would be less than $x^*(T)$, and vice versa.

In addition, the monetary incentive the government will provide at the terminal time T should be at least as much as the profit amount that the firm could potentially generate by dynamically extracting and selling the remaining reserve at instantaneous prices while taking the associated cost into account. From this model, it's possible to determine that amount by solving the integration $\int (p(u)u - c(u))e^{-rt}dt$ from T^* to the time at which the reserve is completely depleted (exhaustion). Any sum of money that exceeds or is equal to this aggregate should provide a competitive incentive to the firm to accept the government's offer and cease extraction efforts without depleting the repository entirely.

Therefore, if the objective of the administration is to conserve as much resource as possible for the research and environmental sustainability, they'd offer larger amounts to the extracting firm. This would reduce the salvage value and allow the firm to reach that level faster (scarcity and, therefore, the marginal contribution of the remaining reserve wouldn't need to climb as much for terminating the operations), resulting in more leftover resources in the wells.

From the firm's viewpoint, if the offered endowment from the government is large enough to compete with their profit prospects at time T , it'd be indifferent between stopping and continuing its extraction activities further, and the moment the endowment exceeded the prospective market returns, it would ideally decide to stop mining and hand the right to the lease over to the government in order to maximize its temporal profits.

Conclusion

The primary objective of this study is to examine the potential impact of adding salvage value and varying discount rates on the optimal extraction of a finite resource (in this case, crude oil) for a monopolist. It is seen from the aforementioned model and numerical analysis that in the presence of a salvage value (such as a lumpsum government payment) big enough to compete with market incentives at time T (As long as the endowment is equal or greater than the possible earnings the firm may generate by selling the remaining resource at the current market price dynamically post- T), the monopolist would discontinue the extraction activities and leave the site with some leftover resources, a decision that will influence its extraction behavior over the time period $[0, T]$. Increasing the discount rate, on the other hand, would discount the future values more rapidly, making present returns more attractive for the monopolist, and hence, the extraction rate would increase and the depletion until the salvage level would take place faster.

However, it is important to acknowledge that this study possesses certain limitations in terms of its scope and methodology. The first obvious shortcoming is that this model assumes particular functional forms for the demand and the cost functions and values for the constants based on previous work without resorting to any empirical data, which makes the model considerably arguable. The second limitation is that the model assumes constant discount rates throughout the lifespan of the extraction, which, in most cases, is not realistic as discount rates tend to vary at different points in time as other investments could become more profitable and accessible (such as a much cheaper renewable source coming online) in comparison during such an extended time period. The third limitation is that this model assumes no replenishment within the time horizon, which might not always hold in a case like the one above where the extraction period extends to more than a century. Last but not least, this method of dynamic optimization could further benefit from having more control and state variables in its model, accounting for more external factors and their interactions. Therefore, it is apparent that there is a great deal of potential for broadening the scope of the study and carrying out additional research using this particular analytical procedure.

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